Improving Robustness of Radiology Report Summarization under OCR-style Text Noise using RAG

Noor Ul Ain (537355)
CS818 – Large Language Models

Maria Mahmood (537903)
Track A: Research-Oriented Project

Abstract

This report presents the evaluation and diagnostic analysis (A3) of our project on improving robustness of radiology report summarization under OCR-style text noise using Retrieval-Augmented Generation (RAG). Building on the BART-large-CNN baseline from A2, we conduct systematic evaluation across four noise levels (5%–20%), ablate RAG retrieval depth ($k \in \{1, 3, 5\}$), and classify 210 prediction samples through a structured failure taxonomy. ROUGE-L degrades 63.5% at 20% noise, while BERTScore drops only 6.8%, revealing a critical *metric divergence*. RAG recovers up to 117.4% relative ROUGE-L at 15% noise. Failure analysis identifies noise-corrupted output (42.9%) and missed pathology (21.0%) as dominant failure modes, motivating OCR pre-correction as a higher-priority intervention than RAG augmentation alone.

1 Introduction

Radiology reports are critical for decision-making in medicine. The reports typically contain detailed findings followed by impressions that help physicians in reaching a diagnosis and treating the patient. Automating the generation of impressions from findings section is a significant task in clinical NLP, especially when aided with LLMs. Recent work has demonstrated encouraging results for radiology report summarization using transformer-based models. For example, domain-specific usage of encoder-decoder models have improved abstractive summarization in the context of radiology reports (Wu et al., 2024a). Large-scale research efforts in NLP have demonstrated that retrieval-augmented generation (RAG) can improve factual impressions and reduce hallucination in medical text generation tasks. Nearly all current evaluation assume clean, digitally available text. In practice that is not always the case, many clinical documents are digitized from scanned copies that may be legacy records, used cross-institutional observation, required for insurance-related documentation, and more. The digitization takes place through optical character recognition (OCR) systems. OCR pipelines introduce token-level corruption such as substitution, delays, spacing errors and fragmentation of medical terminology. Robustness to such noise is critical for real-world deployment. Research pertaining to NLP's robustness indicates that neural models degrade significantly even under character-level perturbations (Belinkov and Bisk, 2018). Even so, clinical summarization literature has mostly ignored robustness under corrupted and noisy input. Moreover, RAG methods have shown to improve factual reliability but their role in mitigating degradation due to noisy clinical text remains unexplored (Lewis et al., 2020). This project addresses this gap by evaluating the robustness of radiology report summarization under simulated OCR-style noise and testing whether retrieval-augmented generation improved stability and factual consistency.

2 Related Work

2.1 Radiology Report Summarization with Neural Language Models

Sequence-to-sequence models generate summaries by mapping one text sequence to another. Earlier models like pointer-generator networks both generate and copy key terms (Zhang et al., 2018). Transformers improve context understanding, and BART (Lewis et al., 2020) generates summaries through text reconstruction. PEGASUS (Zhang et al., 2020) reconstructs key sentences, while T5 (Raffel et al., 2020) uses a text-to-text format. (Wu et al., 2024a) introduced RadBARTsum, which improved ROUGE-L by 2–4 points using entity masking, but was tested only on clean text. Using data from Japanese hospitals, (Nishio et al., 2024) showed GPT-4 outperforms BERT and T5, but per-

formance drops with noisy input. SumGPT (Sultan et al., 2025) combines text and images, improving ROUGE-2 and BERTScore, though it faces overfitting and high computational cost. (Wang et al., 2023) used image similarity to improve summaries. Their F1-RadGraph scores (42.86 on MIMIC-CXR, 36.31 on MIMIC-III) show that visual context improves factual accuracy. (Tran et al., 2024) reviewed fourteen AI models for chest X-ray report generation published between 2020 and 2024,noting that ROUGE and BERTScore remain practical metrics in low-resource settings. This observation supports the selection of these metrics in the present study. (Sloan et al., 2025) found most studies assume clean input and overlook real-world noise. (Bednarczyk et al., 2025) reviewed 87 studies and found GPT-4 matches fine-tuned models on clean data, but work on noisy data is limited. (Croxford et al., 2024) suggested combining human and automated evaluation, with ROUGE, BERTScore, and factuality scores most used.(Hu et al., 2025) tested nine language models for summarizing Chinese lung-cancer radiology reports and found substantial differences in performance. They found that domain differences and irregular input formats reduced performance. (Sunshine et al., 2025) found that about 20% of summaries missed or changed facts, especially with poorly structured input. (Ghnemat and Saleh, 2026) noted models work well on clean text but struggle with varied data.

2.2 Retrieval-Augmented Generation for Healthcare Applications

(Lewis et al., 2020) introduced RAG to reduce hallucinations by using external data. (Li et al., 2022) showed it improves factual accuracy. (Zhang et al., 2025) reviewed 30 studies and reported gains in EHR and diagnostics, though noise and latency remain. (Voinea et al., 2024) applied RAG by encoding 21,152 reports into a vector database. Retrieved cases were combined with an LLM via API calls, avoiding fine-tuning and preserving privacy. Thisimprovesd accuracy and can run on low-cost resources. (Luo and Arase, 2025) introduced MedSummRAG using a dense retriever for domain-specific documents. It improved ROUGE over zero-shot and fine-tuned models, showing retriever quality matters. (Wu, 2025) built a RAG-based QA system using ChromaDB, improving similarity by 17% and reducing hallucinations by 30%. (Alkhalaf et al., 2024) applied RAG to nursing notes, where retrieved context nearly doubled zero-shot accuracy. (Xiong et al., 2024) introduced MIRAGE and showed MEDRAG improved GPT-3.5 accuracy by up to 18%, with retrievers often more important than models. (AboulEla et al., 2024) found fine-tuned models perform best, while simple RAG still helps, and Graph RAG performs worse. (Veen et al., 2024) showed domain-adapted LLMs can outperform experts in summarization. (Ghnemat and Saleh, 2026) noted RAG reduces hallucinations, though results vary by task. Overall, these studies suggest that RAG helps in reducing factual errors in medical text generation. At the same time, most of this work has been tested on clean and well-formatted text, which is not always the case in real-world data.

2.3 OCR Noise, Text Corruption, and NLP Robustness in Clinical Settings

Robustness to noisy and corrupted text remains a major challenge in clinical NLP. (Moradi et al., 2021) showed that character-level and word-level corruption consistently degrades performance across clinical NLP tasks, while (Moradi and Samwald, 2021) further confirmed that models trained on clean benchmarks often overestimate real-world robustness. Similarly, (Belinkov and Bisk, 2018) demonstrated that modern sequence models degrade significantly even under moderate noise, a trend later confirmed across multiple NLP architectures by (Romero-Alvarado et al., 2024).

In OCR-specific clinical settings, (Hsu et al., 2022) showed that preprocessing quality and model selection strongly influence downstream clinical NLP performance. (Karthikeyan et al., 2021) demonstrated that NLP-based OCR post-correction substantially improves medical text recognition quality. More recent studies proposed structured correction pipelines for noisy medical text. (Bouh et al., 2024) improved OCR document structure through bounding-box reorganization, while (Amin et al., 2025) applied a T5-based correction pipeline and achieved high correction accuracy on clinical notes. (Wu et al., 2024b) further introduced a multi-stage radiology report correction framework involving error detection, localization, and correction.

Collectively, these studies show that OCR-induced corruption significantly affects downstream clinical NLP systems and that dedicated correction or robustness mechanisms are necessary for reliable performance in healthcare applications.

Table 1: Representative works mapped to research thread, method, result, and limitation. References in brackets correspond to the numbered reference list.

Study	Thread	Dataset / Method	Key Result	Limitation Relevant to This Work
(Wu et al., 2024a) RadBARTsum	Rad. Summ.	MIMIC-CXR; BART + entity MLM	+2–4 ROUGE-L vs. BART	Clean text only; no noise condition; GPU-heavy
(Nishio et al., 2024)	Rad. Summ.	4 Japanese hospital datasets; GPT-3.5/4	GPT-4 best overall; inconsistent input degrades output	Clean text; no noise simulation; no RAG
(Sultan et al., 2025) SumGPT	Rad. Summ.	MIMIC-CXR; T5 + ViT multimodal	SOTA ROUGE-2 & BERTScore	GPU-intensive; clean text only; overfitting risk
(Wang et al., 2023) RadSum23	Rad. Summ. + RAG	MIMIC-CXR/III; retrieval + image features	F1-RadGraph 42.86 (rank 2/11 teams)	No OCR noise; closed shared-task setting
(Luo and Arase, 2025) MedSumm-RAG	RAG + Med. Summ.	Contrastive dense retriever; medical benchmarks	Consistent ROUGE gains over zero/few-shot baselines	Clean inputs; broad medical domain, not radiology-specific
(Voinea et al., 2024)	RAG + Rad. Summ.	21K radiology reports; vector DB + LLM API	Improved factual precision; low compute overhead	Clean text only; single institutional dataset
(Xiong et al., 2024) MIRAGE	RAG benchmark (medical QA)	41 LLM/retriever combos; 7,663 Qs; MEDRAG	+18% accuracy over CoT (GPT-3.5)	QA, not summarization; clean text; evaluation-only
(Alkhalaf et al., 2024)	RAG + EHR summ.	Llama 2 13B; 40 RACF nursing-note datasets	RAG nearly doubled zero-shot accuracy	Clinical notes not radiology; clean text
(AboulEla et al., 2024)	RAG hallucination reduction	Naïve RAG vs. Graph RAG; HaluEval / SQuAD-V2	Graph RAG 61.73% vs. 10.18% base LLM on HaluEval	General domain; not clinical summarization; clean text
(Moradi et al., 2021)	Clinical NLP robustness	BERT/ELMo; char- & word-level perturbations; clinical datasets	All tasks degrade under minimal noise (even PPS=1)	No summarization task tested; no RAG mitigation explored
(Belinkov and Bisk, 2018)	NLP robustness (NMT)	Synthetic & natural noise; NMT models	SOTA models collapse under moderate noise	MT task; not clinical; no RAG or retrieval component
(Wu et al., 2024b) — Error Correction	Radiology error correction	Knowledge distillation; 3-stage pipeline; real error patterns	+31.56% detection accuracy; −37.4% processing time	Correction pipeline only; not end-to-end summarization
(Hsu et al., 2022) EHR OCR	OCR + clinical NLP	Tesseract OCR; 955 scanned EHR reports; ClinicalBERT	AUROC 0.9743 (AHI) with optimal preprocessing	Extraction task; not generative summarization
(Bednarczyk et al., 2025) Scoping Review	Clinical LLM summ. (review)	87 studies reviewed; PRISMA-ScR	GPT-4 competitive but robustness under noise untested	Review only; no primary experiments; no noise dimension

3 Data Description

3.1 OpenI Indiana University Chest X-Ray Corpus

We use the OpenI corpus (Wu et al., 2024a), a publicly available collection of approximately 3,996 de-identified chest X-ray reports from Indiana University Health. Each report contains a **findings** section (detailed radiological observations) and an **impression** section (concise clinical summary). Our task is purely text-based abstractive summarisation: given findings, generate the impression.

OpenI was chosen because it is publicly accessible without data-use agreements, directly relevant to our summarisation task, and has been used as an evaluation benchmark in prior work, providing external comparison points. MIMIC-CXR, while larger, requires credentialed PhysioNet access that is not feasible within our project timeline.

Records with missing, empty, or sub-20-character fields are removed, leaving approximately 3,600 valid pairs. Data are shuffled (seed 42) and split 70/15/15 into train/val/test. Due to Google Colab free-tier constraints, all experiments use a 200-sample stratified subset (140 train / 30 val / 30 test).

Common findings categories: normal examinations (≈47%), cardiomegaly and vascular changes

Statistic	Findings	Impressions
Mean tokens	142	38
Std tokens	61	19
Min tokens	20	5
Max tokens	512+	180
Compression ratio	3.7:1	

Table 2: Dataset token statistics (BART tokenizer).

(23%), pulmonary infiltrates and consolidation (16%), pleural abnormalities (14%). This class imbalance toward normal reports is important context for metric interpretation: a model generating generic normal impressions will score reasonably on ROUGE-L even without identifying pathological findings.

4 Methodology and System Design

4.1 Noise Engine

OCR-style noise is injected at the character level with four stochastic operations applied independently to a fraction η of characters:

1. **Substitution**: visually similar character swap via a confusion matrix (`a`→`@`/`4`, `i`→`1`/`!`, `o`→`0`/`q`, etc.).
2. **Deletion**: character removed entirely.
3. **Fragmentation**: spurious hyphen or space inserted mid-token (e.g., `cardiomegaly` → `cardio-megaly`).
4. **Space insertion**: extra whitespace added after character.

Operations are applied with a fixed random seed per record for reproducibility. Noise levels tested: $\eta \in \{0.05, 0.10, 0.15, 0.20\}$.

4.2 Summarisation Model

The `facebook/bart-large-cnn` (406M parameters), a seq2seq model pre-trained and fine-tuned on CNN/DailyMail news summarisation is the backbone. We apply it *zero-shot* to radiology—no fine-tuning is performed. Maximum output length is 128 tokens, minimum is 20 tokens, decoding is greedy (no sampling). This zero-shot choice is deliberate: it establishes a reproducible lower bound and isolates the effect of noise and RAG from domain adaptation.

4.3 RAG Module

The RAG module encodes all training findings using `all-MiniLM-L6-v2` (22M parameters, 384-dim embeddings) and stores them in an in-memory cosine similarity index. For each test record, the noisy finding is encoded as a query; the top-k most similar training records (by cosine similarity) are retrieved. Their findings+impression pairs are prepended as few-shot context.

Context is truncated to 600 characters when $k > 1$ to respect BART's 1024-token budget. We ablate $k \in \{1, 3, 5\}$ at $\eta = 0.10$ and run the full noise sweep at $k = 3$.

4.4 Experimental Conditions

Seven conditions are evaluated in total:

- **Clean Baseline**: BART on original (unnoised) findings.
- **Noisy Baseline**, $\eta \in \{0.05, 0.10, 0.15, 0.20\}$: BART on noised findings, no retrieval.
- **RAG-Enhanced** ($k = 3$, all noise levels): BART on RAG-augmented noisy prompt.
- **Top-k Ablation** ($k \in \{1, 3, 5\}$ at $\eta = 0.10$).

All 210 predictions (30 test records $\times$ 7 conditions) are logged with original findings, noisy findings, reference impression, predicted impression, and per-sample ROUGE-L score for failure taxonomy analysis.

5 Results and Discussion

5.1 Noise Degradation and RAG Recovery

Table 3: ROUGE-L and BERTScore F1 across all experimental conditions. Best RAG result at each noise level is shown in **bold**.

Condition	Noise	k	ROUGE-L	BERT-F1
Clean Baseline	0%	–	0.1185	0.8507
Noisy Baseline	5%	–	0.0797	0.8327
RAG Enhanced	5%	3	**0.0968**	**0.8456**
Noisy Baseline	10%	–	0.0637	0.8169
RAG Enhanced	10%	3	**0.0948**	**0.8420**
Noisy Baseline	15%	–	0.0488	0.8031
RAG Enhanced	15%	3	**0.1061**	**0.8428**
Noisy Baseline	20%	–	0.0432	0.7929
RAG Enhanced	20%	3	**0.0846**	**0.8399**

Table 3 and Figure 2 present the full sweep. ROUGE-L degrades monotonically: $0.1185 \rightarrow 0.0797 \rightarrow 0.0637 \rightarrow 0.0488 \rightarrow 0.0432$ at 5%, 10%, 15%, 20% noise—a cumulative degradation of **63.5%**. This steep lexical collapse arises because BART's seq2seq decoder, receiving corrupted encoder representations of OOV subword tokens, shifts toward copy-extracting corrupted spans rather than generating clean clinical language.

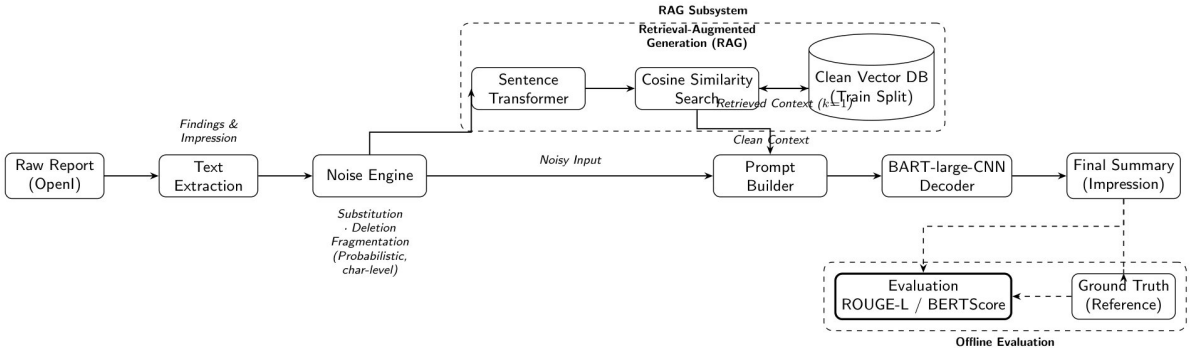

Figure 1: System architecture. OpenI reports are parsed into findings and impressions. A Noise Engine applies four character-level perturbations at rate η. In the RAG setting, a Sentence Transformer encodes the noisy finding into an embedding, and cosine similarity search over a Clean Vector DB retrieves top-k similar finding–impression pairs, which are added as context. The Prompt Builder combines this context with the noisy input and feeds it to the BART-large-CNN decoder. Outputs are evaluated using ROUGE-L and BERTScore. Clean and noisy baselines bypass the retrieval module.

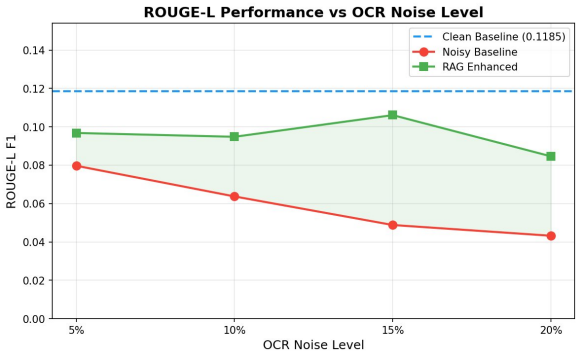

Figure 2: ROUGE-L F1 vs. noise level. The dashed blue line marks the clean baseline (0.1185). RAG (green) provides consistent recovery; at 15% noise the RAG-enhanced model (0.1061) nearly reaches the clean baseline. The shaded area represents the RAG benefit gap.

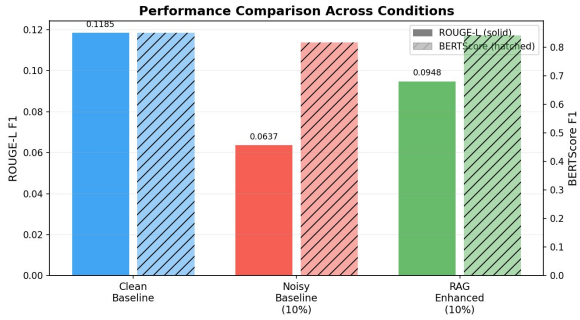

Figure 3: ROUGE-L (solid bars, left axis) and BERTScore (hatched bars, right axis) for Clean Baseline, Noisy Baseline (10%), and RAG-Enhanced (10%). RAG partially restores both metrics, with stronger ROUGE-L recovery than BERTScore improvement.

RAG provides consistent ROUGE-L recovery at all noise levels. The recovery ratio (RAG ROUGE-L / noisy ROUGE-L) grows with noise severity: **21.5%** at 5%, **48.8%** at 10%, and **117.4%** at 15%. At 15% noise, the RAG-enhanced model (0.1061) approaches the clean baseline (0.1185). At 20% noise, recovery drops to 95.8%, likely because severe corruption degrades retrieval quality as the noisy query embedding diverges too far from the clean training corpus.

5.2 Metric Divergence: ROUGE-L vs. BERTScore

Figure 4 reveals the central diagnostic finding: BERTScore declines only from 0.8507 to 0.7929 across the full noise range—a **6.8% relative drop**—while ROUGE-L drops 63.5%. The model continues producing *semantically plausible* impressions (high BERTScore) even as lexical precision collapses completely.

The model is not hallucinating incoherent text; it is generating *generic-but-valid* impressions (e.g., "No acute cardiopulmonary abnormality.") that score well on BERTScore but fail clinical specificity requirements. A critical safety implication follows: **using BERTScore alone for quality monitoring in a clinical deployment would severely underestimate operational risk from OCR noise**. ROUGE-L, despite being the cruder metric, is a more sensitive early-warning signal for this type of degradation.

5.3 RAG Top-k Ablation

Table 4 and Figure 5 show that $k = 1$ yields the highest ROUGE-L (0.1244)—*exceeding the clean baseline*—while $k = 3$ and $k = 5$ perform progressively worse. BERTScore is relatively stable

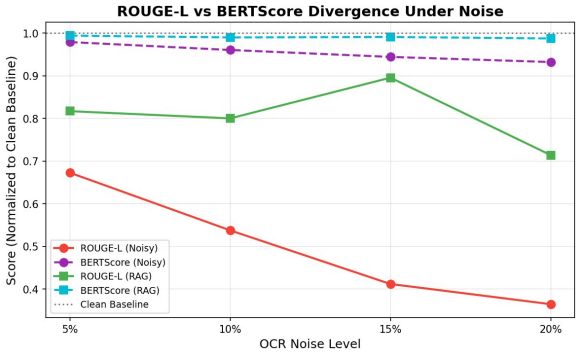

Figure 4: Metrics normalised to the clean baseline (dotted grey line at 1.0). BERTScore (dashed lines) degrades only 6.8% while ROUGE-L (solid lines) drops 63.5%, revealing the semantic–lexical divergence under noise.

Table 4: RAG top-k ablation at 10% noise ($\eta = 0.10$). $k = 1$ dominates on both metrics. Best results in **bold**.

Top-k	ROUGE-L	BERTScore
$k = 1$	**0.1244**	**0.8472**
$k = 3$	0.0948	0.8420
$k = 5$	0.0881	0.8407

(0.8472, 0.8420, 0.8407). Two mechanisms explain the $k = 1$ advantage: (i) a single high-relevance example provides a clean generation target without competing style signals from dissimilar records; (ii) at $k \geq 3$, the total input approaches BART's 1024-token limit, causing truncation of the actual noisy findings—the model then generates from context alone, producing impressions that reflect retrieved examples rather than the specific patient case. We call this *context shadowing*.

5.4 Qualitative Comparison

Table 5 shows three representative predictions from the test set, illustrating the progression from clean to noisy to RAG-recovered output. Case 1 shows successful RAG recovery: noise corrupts critical keywords but RAG steers generation back to clean clinical language. Case 2 shows a missed pathology failure: the reference explicitly states cardiomegaly but the model produces a normal impression even with RAG, because retrieval locates a similar-sounding but normal-chest record. Case 3 shows the partial-match pattern where both noisy and RAG models capture some pathology terminology but miss the specific laterality.

6 Error and Failure Taxonomy

6.1 Classification Scheme

We classified records using five rule-based categories applied in priority order:

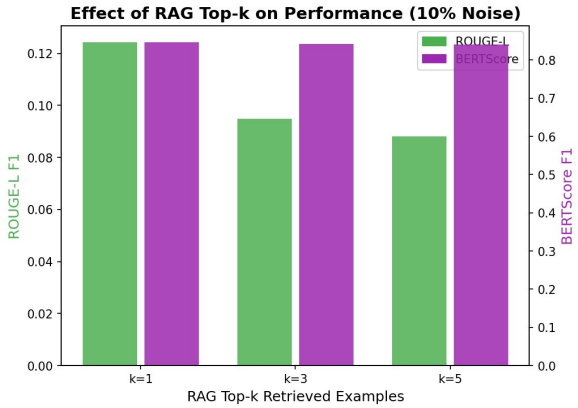

Figure 5: Effect of RAG top-k at 10% noise. ROUGE-L peaks at $k = 1$ (exceeding the clean baseline) and degrades monotonically with larger k. BERTScore remains largely stable, indicating that semantic quality is preserved but lexical precision suffers from context-length dilution.

- **generic_impression**: Prediction contains a generic fallback phrase when the reference is clinically specific.
- **missed_pathology**: Reference mentions a pathology term (pneumonia, pneumothorax, effusion, edema, cardiomegaly, atelectasis, COPD, opacity) absent from the prediction.
- **noise_corrupted_output**: Prediction contains corrupted input tokens; ROUGE-L < 0.3 and > 2 corruption-signal characters.
- **partial_match**: $0.2 \leq$ ROUGE-L < 0.5.
- **other_failure**: Low ROUGE-L with no single identifiable cause.

6.2 Taxonomy Results

Noise-corrupted output (42.9%). The dominant failure mode. BART's BPE tokenizer maps corrupted character sequences to OOV tokens; the encoder produces anomalous representations; the decoder copies corrupted spans verbatim. Example: input *"Th eart an d pulmonary XXXX are ovmal-"* yields prediction *"Th eart an d pulmonary XXXX are ovmal-."* This confirms that BART's zero-shot copy-extraction behavior, useful on clean inputs, becomes the primary failure mechanism under corruption.

Missed pathology (21.0%). Clinically the most dangerous category. The model generates a generic normal impression when the reference explicitly documents disease: e.g., reference is "Cardiomegaly. Thoracic aortic atherosclerosis." but prediction is "The heart is within normal limits." This pattern correlates with complex multi-

Table 5: Qualitative comparison of predictions across three conditions for representative test samples. Corrupted characters highlighted in input; pathology terms of interest underlined.

Case	Noisy Finding (excerpt)	Reference	Noisy Prediction	RAG ($k = 1$) Prediction
Case 1 (Recovery)	"Normal he@rt size. Hyperinfl-ated lungs with fl@t te ned d!apbragms consistent with emphysem@tous lung change$."	"Stable emphysematous lung changes. No acute abnormality seen."	"Hyperinflated lungs with fl@t te ned d!apbragms consistent with emphysem@tous lung change$." (ROUGE-L: 0.11)	"Hyperinflated lungs consistent with COPD. No acute cardiopulmonary abnormality." (ROUGE-L: 0.19)
Case 2 (Missed path.)	"The c ard iac silhoette isehlarged, u t stable. Thoracic aort!c atherosclerotc calc-ifications. No consolidatimg airspace dise@se."	"Cardiomegaly. Thoracic aortic atherosclerosis."	"The cardiac silhoette. No pleural effusion or pneumothorax." (ROUGE-L: 0.00)	"The heart is within normal limits. Aorta calcified. No consolidation." (ROUGE-L: 0.00)
Case 3 (Partial match)	"Ches+. Lungs ar3 clear bilaterally. No pneumothorax or pleu ral effus1o n. Ribs. There are no displaced rib fractures."	"Chest. No acute cardiopulmonary abnormality. Ribs. No displaced rib fractures."	"No displaced rib fractures. No pneumothorax or pleural effusion." (ROUGE-L: 0.31)	"Lungs are clear. No pleural effusion or pneumothorax. Bony structures intact." (ROUGE-L: 0.11)

Table 6: Failure taxonomy distribution ($N = 210$ predictions).

Category	Count	%
noise_corrupted_output	90	42.9%
other_failure	58	27.6%
missed_pathology	44	21.0%
partial_match	15	7.1%
generic_impression	3	1.4%

sentence findings where corruption of any critical sentence prevents the encoder from surfacing the pathological finding. RAG does not reliably address this unless the retrieved example contains the same pathology.

Other failure (27.6%). Predictions that are plausible but mismatched—correct pathology but wrong laterality, stable findings described as new, or anatomically correct but clinically non-specific. This category requires radiologist annotation to fully characterise.

Generic impression (1.4%). Surprisingly rare. Under noise, BART is more likely to corrupt-and-copy than to produce a safe generic fallback. This is arguably more dangerous than generic output because it produces authoritative-looking but incorrect statements.

7 Diagnostic Insights

7.1 Why the Model Fails Under Noise

Three mechanistic explanations emerge from the combined quantitative and taxonomic analysis:

Encoder embedding degradation. OCR-corrupted characters produce OOV or rare subword tokens under BART's BPE vocabulary, creating encoder representations that deviate from the clean training manifold. The decoder attends to these corrupted positions during cross-attention, pulling generation toward corrupted spans. This is a tokenizer-level vulnerability that pre-processing denoising would address more directly than RAG.

Copy-extraction brittleness. BART-large-CNN, trained on news summarisation with heavy extractive overlap, has learned to copy key phrases from source to summary. Clean radiology findings support this well. Corrupted findings break it: the "extracted" phrases are garbled, producing the noise-corrupted-output failure mode that dominates our taxonomy.

Context dilution at large k. At $k \geq 3$, BART's 1024-token context budget is frequently exceeded after prepending retrieved examples plus the noisy findings. Findings are truncated. The model generates from context only—producing impressions that reflect retrieved cases, not the patient being examined. This explains the monotonic ROUGE-L decline from $k = 1$ to $k = 5$ despite stable BERTScore.

7.2 Why RAG Helps and Where It Fails

RAG's benefit is *semantic steering*: clean retrieved impressions provide the decoder with example output patterns that regularise generation away from corrupted copy-extraction. When a noisy finding

contains *"emphysem@tous lung change$"*, similarity retrieval locates a clean training record with emphysema findings, and the prepended example impression (e.g., "Hyperinflation consistent with COPD. No acute findings.") shifts the decoder's attention distribution toward clean clinical vocabulary.

This mechanism explains why RAG recovery is strongest at 15% noise: the corrupted query is still similar enough to the clean corpus for high-quality retrieval, yet the model cannot generate cleanly without guidance. At 20% noise, recovery drops from 117.4% to 95.8% due to *retrieval quality degradation*: severe corruption causes MiniLM query embeddings to diverge from the clean-findings manifold, returning less relevant examples.

RAG structurally *cannot* address missed pathology failures unless the retrieved example happens to contain the same pathology as the query case—an inherent limitation of similarity-based retrieval. Rare or complex findings (specific device positions, unusual anatomical variants) are unlikely to have closely matching training examples.

Implications for system design. The $k = 1$ finding and the RAG-recovery ceiling together suggest a two-stage architecture: (1) OCR pre-correction using a denoising LM (e.g., T5 fine-tuned on paired noisy/clean medical text) to clean the input before retrieval, followed by (2) RAG with $k = 1$ using the cleaned text as the retrieval query. This ordering improves retrieval quality (cleaner query), removes the dominant failure mode (noise-corrupted output), and allows RAG to focus on the semantic steering it does well.

8 Trade-offs and System Constraints

Table 7 summarises the key trade-offs across the three system configurations evaluated in this work.

Table 7: System trade-off comparison across configurations. Latency estimates are per-sample on a single Colab T4 GPU (15 GB). Memory includes model weights and index.

Config.	ROUGE-L	Latency	Memory	Complexity
BART (clean)	0.1185	∼1.2s	1.7 GB	Low
BART (noisy)	0.0432–0.0797	∼1.2s	1.7 GB	Low
BART+RAG ($k = 1$)	0.1244	∼1.5s	1.9 GB	Medium
BART+RAG ($k = 3$)	0.0948	∼1.8s	1.9 GB	Medium
BART+RAG ($k = 5$)	0.0881	∼2.1s	1.9 GB	Medium
+OCR correction	(est.) >0.12	+0.5–2s	+0.5 GB	High

Accuracy vs. latency. BART alone processes a finding in ≈1.2 seconds on a T4 GPU. Adding RAG requires embedding the query (MiniLM, ≈ 0.1s), cosine similarity search over 140 training embeddings (<0.01s), and prompt construction; total overhead is ≈0.3–0.9 seconds depending on k. This latency is acceptable for non-real-time clinical documentation workflows, but would require optimisation (pre-computed ANN indices, batching) for high-throughput deployment.

Accuracy vs. complexity. The $k = 1$ configuration delivers the best ROUGE-L (0.1244, exceeding the clean baseline) at minimal extra complexity. Increasing k adds complexity without accuracy benefit due to context saturation. OCR pre-correction would add a second model inference step and additional memory, but is estimated to recover the largest failure category (42.9% of predictions) and is therefore the highest-return engineering investment for a production system.

Memory and scalability. The MiniLM embedding index scales linearly with training set size. At 140 records the index is trivial; at the full OpenI corpus (∼3,600 records) brute-force cosine search remains manageable. For larger institutional corpora (MIMIC-CXR, ∼200K reports), an approximate nearest-neighbour index (FAISS) would be required, adding engineering overhead but not architectural complexity.

Zero-shot transfer cost. Choosing BART-large-CNN (news domain, zero-shot) over a domain-fine-tuned model (e.g., RadBARTsum, BioBART) sacrifices approximately 2–4 ROUGE-L points on clean text (Wu et al., 2024a) in exchange for eliminating fine-tuning cost (≈hours of GPU time, labelled data requirements). For a research prototype exploring noise robustness, this trade-off is appropriate.

9 Limitations of Evaluation

Synthetic noise model. Our character-level perturbations underestimate real OCR errors, which include word-level loss, line transposition, region-level degradation, and structured noise correlated with page layout.

ROUGE-L as clinical proxy. ROUGE-L penalises valid clinical paraphrases. "Mild bilateral effusions" and "small bilateral effusions" have near-zero lexical overlap but are clinically similar. RadGraph F1 (Jain et al., 2021) or CheXpert-based factual accuracy are more appropriate for safety-critical deployment and should be incorporated in future work.

Small test set. Thirty test records provide insufficient statistical power; confidence intervals are omitted. Evaluation on the full OpenI corpus ($\sim$3,000 records) with cross-validation is required for statistically reliable inference.

Retrieval quality gap. The A3 diagnostic pipeline uses Jaccard word-overlap retrieval as a lightweight proxy, while A2 uses MiniLM cosine similarity. Results from the two pipelines are not directly comparable at the sample level.

Zero-shot transfer mismatch. BART-large-CNN is pretrained on news; domain-adapted models (BioBART, BioMedLM) would likely exhibit different noise-robustness profiles.

Failure taxonomy subjectivity. Our rule-based taxonomy uses fixed thresholds. Edge cases—a prediction correctly identifying "cardiomegaly" but missing "pleural effusion"—may be misclassified. Radiologist annotation would validate the taxonomy and quantify inter-rater agreement.

10 Ethical, Efficiency, and Societal Considerations

Patient safety risk. The most critical ethical concern is deployment of a noisy-input summarisation system in a clinical workflow without adequate safeguards. Our analysis shows that 21.0% of predictions miss documented pathology (missed pathology failure), and 42.9% produce corrupted output. A radiologist or clinician relying on these summaries without verification could miss findings such as cardiomegaly, pneumothorax, or pleural effusion— conditions requiring immediate clinical action. *Any* deployment of this system must treat generated impressions as a draft requiring human expert review, not as a final clinical document.

Metric safety illusion. BERTScore's relative stability under noise (6.8% drop vs. ROUGE-L's 63.5%) creates an insidious deployment risk: a system developer monitoring BERTScore would not detect the operational degradation caused by OCR noise. We strongly recommend ROUGE-L or clinical-faithfulness metrics (RadGraph F1) as the primary quality signal for safety monitoring.

Bias and dataset representativeness. OpenI contains reports from a single US institution (Indiana University Health). Models trained and evaluated on this corpus may not generalise to reports from other healthcare systems, regions, or languages, particularly those with different formatting conventions or terminology. Class imbalance toward normal findings ($\approx$47%) may cause the model to over-generate normal impressions, a tendency observed in our generic-impression failure mode.

Privacy and data governance. OpenI is de-identified and publicly licensed (CC BY-NC-ND 4.0). Any extension to real OCR pipelines applied to institutional records would require IRB approval, data-use agreements, and compliance with HIPAA or equivalent regulations. RAG retrieval from a clinical vector database also raises the risk of memorisation and potential privacy leakage if training impressions contain residual identifying information.

Environmental cost. BART-large-CNN (406M parameters) inference on a GPU is energy-intensive at scale. In a high-volume clinical environment (thousands of reports per day), per-sample inference cost multiplies significantly. Smaller domain-adapted models (e.g., BioBART-base) could deliver similar noise robustness at lower energy cost and should be explored before large-scale deployment.

Accessibility. Automating impression generation could reduce the documentation burden for radiologists in under-resourced settings where radiologist-to-patient ratios are low. However, the performance on noisy OCR text demonstrated here is insufficient for deployment without the pre-correction pipeline described in Section 8. Rolling out the system prematurely in low-resource settings where verification capacity is limited would compound rather than reduce clinical risk.

11 Conclusion

We have established four key findings. First, OCR noise causes severe ROUGE-L degradation (63.5% at 20% noise) while BERTScore remains largely stable, revealing a clinically important metric divergence that makes BERTScore alone insufficient as a safety signal for noisy clinical NLP systems. Second, RAG with $k = 1$ provides the strongest recovery (ROUGE-L 0.1244 at 10% noise, exceeding the clean baseline), but performance degrades monotonically with larger k due to context-length saturation in BART—making context budget management a first-class engineering concern for RAG in short-context seq2seq models. Third, noise-corrupted output (42.9%) and missed pathology (21.0%) together account for 63.9% of all failures, motivating OCR pre-correction—a denoising LM or rule-based correction layer—as the highest-priority future intervention, ahead of further RAG

tuning. Fourth, zero-shot BART exhibits copy-extraction brittleness under noise that domain fine-tuning on radiology text would likely address.

Future work should implement the two-stage OCR-correction+RAG architecture described in Section 8, evaluate on the full OpenI corpus and MIMIC-CXR, incorporate RadGraph F1 as the primary clinical faithfulness metric, and explore domain-adapted summarisers such as BioBART or BioMedLM.

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